

Analysis General Exam, January 2010

January 2010

Closed book, closed notes. Please pledge. In each problem, justify all assertions, show calculations, and identify those theorems which you invoke in your arguments.

Problem 1

Let $w_1, w_2, \dots, w_n$ be n points on the unit circle $\mathbb{T} \equiv \{z \in \mathbb{C} : |z| = 1\}$. Show that there exists a point $z \in \mathbb{T}$ so that the product of the distances from z to w_j (i.e., $\prod_{j=1}^n |z - w_j|$) is at least 1. Then show there is a point $v \in \mathbb{T}$ so that the product of the distances from v to the points w_j is exactly 1.

Problem 2

Suppose that f is analytic in an open set containing the closed unit disk $\overline{\mathbb{D}} = \{z : |z| \leq 1\}$, except for a simple pole at z_0 where $|z_0| = 1$. Show that if

$$\sum_{n=0}^{\infty} a_n z^n$$

is the power series for f in $\mathbb{D}$, then

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n+1}} = z_0$$

Problem 3

Suppose that f and g are analytic in an open set containing the closed unit disk $\overline{\mathbb{D}} = \{z \in \mathbb{C} : |z| \leq 1\}$. Suppose that f has a simple zero at $z = 0$ and no other zero in $\overline{\mathbb{D}}$. Set

$$f_\epsilon(z) = f(z) + \epsilon g(z)$$

Show that if ϵ is sufficiently small, then f_ϵ has a unique zero in $\overline{\mathbb{D}}$.

Problem 4

Evaluate

$$\int_0^\infty \frac{dx}{1+x^n}$$

for $n = 2, 3, 4, \dots$ by integrating over the boundary of an appropriately chosen "pie-shaped" region.

Problem 5

Let $\mathcal{B}(\mathbb{R}^1)$ be the collection of Borel sets on the real line and $\mathcal{B}(\mathbb{R}^2)$ those of the plane.

(a) Show that sections of Borel sets in the plane, e.g., sets of the form $B_y = \{x \in \mathbb{R} : (x, y) \in B\}$ with $B \in \mathcal{B}(\mathbb{R}^2)$ are in $\mathcal{B}(\mathbb{R}^1)$.

(b) Suppose that $f(x, y)$ is a Borel measurable function on the plane, $\mathbb{R}^2$. Show that for fixed y , $f_y(x) = f(x, y)$ is a Borel function on $\mathbb{R}$.

Problem 6

Let $0 < a < 1 < b$ be real constants. Define the function $S(x)$ on $[0, \infty)$ by the equation

$$S(x) = \sum_{n=1}^{\infty} a^n \chi_{[0, b^n]}(x)$$

where $\chi_{[0, b^n]}(x)$ is the characteristic function for $[0, b^n]$, equal to 1 on this interval and zero otherwise.

(a) Show that $S(x) \in L^p([0, \infty), dx)$, i.e., is L^p -integrable (with Lebesgue measure), provided $ab^{1/p} < 1$.

(b) Show that

$$S(x) \leq \frac{x^{(\ln a / \ln b)}}{1 - a}$$

Problem 7

Let $\{\phi_n\}$ be a complete orthonormal set of functions in the real Hilbert space $L^2([0, 1])$ with inner product

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx$$

For $0 \leq a \leq 1$, set

$$h(x) = \sum_{n=1}^{\infty} \phi_n(x) \int_0^a \phi_n(y)dy$$

(a) Show that the series converges in an L^2 -sense to an L^2 function.

(b) Show that for any $b, 0 \leq b \leq 1$,

$$\int_0^b h(x)dx = \sum_{n=1}^{\infty} \int_0^b \phi_n(x)dx \int_0^a \phi_n(y)dy$$

(c) Show that

$$\int_0^b h(x)dx = \min(a, b)$$